

A Numerical Audit of Fractional-Derivative Libraries

Verifying `differint`, `pyfod`, and `numfracpy` against an independent Wolfram reference

1. Introduction to the fractional calculus used

Classical differentiation is defined only for integer orders. **Fractional calculus** extends the derivative and the integral to arbitrary real (or complex) orders, producing a family of operators D^α and I^α that interpolate between the identity ($\alpha = 0$), the ordinary derivative ($\alpha = 1$), and the ordinary integral ($\alpha = -1$). The motivation is both theoretical and practical: fractional operators encode **memory** and **long-range dependence** — the value of $D^\alpha f$ at a point depends on the whole history of f through a weakly singular integral kernel — which makes them natural models for viscoelasticity, anomalous diffusion, subdiffusion, and a growing class of financial and control problems.

Three definitions of the fractional derivative are in common use, and they are the ones exercised in this experiment. All are taken with lower terminal 0.

Riemann-Liouville fractional integral. For $\alpha > 0$,

$$I^\alpha f(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt.$$

Riemann-Liouville fractional derivative. For $n-1 < \alpha < n$ with $n = \lceil \alpha \rceil$,

$$D_{RL}^\alpha f(x) = \frac{d^n}{dx^n} I^{n-\alpha} f(x) = \frac{1}{\Gamma(n-\alpha)} \frac{d^n}{dx^n} \int_0^x (x-t)^{n-\alpha-1} f(t) dt.$$

For the half-order case $0 < \alpha < 1$ this reduces to

$$D_{RL}^\alpha f(x) = \frac{1}{\Gamma(1-\alpha)} \frac{d}{dx} \int_0^x \frac{f(t)}{(x-t)^\alpha} dt.$$

Caputo fractional derivative. This moves the ordinary derivative *inside* the integral:

$$D_C^\alpha f(x) = I^{n-\alpha} f^{(n)}(x) = \frac{1}{\Gamma(n-\alpha)} \int_0^x (x-t)^{n-\alpha-1} f^{(n)}(t) dt,$$

so for $0 < \alpha < 1$,

$$D_C^\alpha f(x) = \frac{1}{\Gamma(1-\alpha)} \int_0^x \frac{f'(t)}{(x-t)^\alpha} dt.$$

The two are related by

$$D_C^\alpha f(x) = D_{RL}^\alpha f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(0)}{\Gamma(\alpha-k)} x^{\alpha-k-1},$$

which means that whenever the relevant initial derivatives vanish (in particular $f(0) = 0$, as for $f = \sqrt{x}$), the Caputo and Riemann-Liouville derivatives coincide for $0 < \alpha < 1$. The

Caputo form is preferred in initial-value problems because it requires only classical initial conditions, but it **demands** $f \in C^n$ — a regularity that $\sqrt{x}$ does not satisfy for $\alpha \geq 1$.

Grünwald-Letnikov derivative. This is the fully discrete, directly implementable definition:

$$D_{GL}^\alpha f(x) = \lim_{h \rightarrow 0} h^{-\alpha} \sum_{m=0}^{\lfloor x/h \rfloor} (-1)^m \binom{\alpha}{m} f(x - mh), \quad \binom{\alpha}{m} = \frac{\alpha(\alpha-1)\cdots(\alpha-m+1)}{m!}.$$

For a large class of functions the Grünwald-Letnikov sum converges to the Riemann-Liouville derivative, and because it is a simple weighted sum it is the workhorse of numerical implementation.

The power-function rule. The entire verification strategy rests on a single closed form. For $\beta > -1$ and $0 < \alpha < \beta + 1$,

$$D_{RL}^\alpha x^\beta = \frac{\Gamma(\beta+1)}{\Gamma(\beta+1-\alpha)} x^{\beta-\alpha}.$$

The null-space phenomenon. This formula silently collapses when $\Gamma(\beta+1-\alpha)$ has a pole, i.e. when $\beta+1-\alpha \in \{0, -1, -2, \dots\}$. In that case the coefficient is exactly zero and

$$D_{RL}^\alpha x^\beta = 0.$$

This is not a numerical accident but a genuine structural property: powers of x whose exponent falls below $\alpha - 1$ are annihilated by the Riemann-Liouville derivative. This phenomenon is the entire reason the second test in this experiment has a *zero* target.

2. Experimental design

Two test problems were run, both with the same function $f(x) = \sqrt{x}$ evaluated at $x = 0.5$, differing only in the order of differentiation:

- **T1 — half-order.** $\alpha = \frac{1}{2}$. By the power rule with $\beta = \frac{1}{2}$, $D_{RL}^{1/2} \sqrt{x} = \frac{\Gamma(3/2)}{\Gamma(1)} x^0 = \frac{\sqrt{\pi}}{2}$, a nonzero constant.
- **T2 — order three-halves.** $\alpha = \frac{3}{2}$. Here $\beta+1-\alpha = \frac{1}{2} + 1 - \frac{3}{2} = 0$, so $D_{RL}^{3/2} \sqrt{x} = 0$ exactly (the null-space case).

The three libraries under test — **differint**, **pyfod**, and **numfracpy** — were each installed in a Python 3.14.3 virtual environment. An independent verification engine, **WolframScript** (Mathematica kernel 14.2.1), was used as the reference. For every test the reference was computed four independent ways: (i) the closed power-function form, (ii) a symbolic Riemann-Liouville evaluation (inner integral, then ordinary differentiation), (iii) a purely numeric Riemann-Liouville evaluation (central differences of the inner integral), and (iv) a direct Grünwald-Letnikov sum. Agreement among these four routes establishes the target; the libraries are then measured against it.

The error metric differs by test. In T1 the target is nonzero, so **relative** error is used. In T2 the target is zero, where relative error is undefined, so **absolute** error is used.

A practical note on the reference tool: this WolframScript build's `--input <file>` flag silently no-ops (the kernel starts but never executes the file), whereas `--code "$(cat <file>)"` works. All reference computations were therefore driven through `--code`.

3. Test 1 — half-order derivative, $\alpha = \frac{1}{2}$

Derivation. With $\beta = \alpha = \frac{1}{2}$,

$$D_{RL}^{1/2} \sqrt{x} = \frac{\Gamma(3/2)}{\Gamma(1)} x^0 = \frac{\sqrt{\pi}}{2} \approx 0.886226925452758.$$

Wolfram reference. All four routes agree to 20 digits:

Reference route	Value
Closed form	0.8862269254527580136490837416705725914
<i>SymbolicR</i> – <i>L</i> ($I(x) = \frac{\pi x}{2}$)	0.8862269254527580136490837416705725914
Numeric R–L (central difference of I)	0.8862269254527580136490836911567791782
Numeric Caputo	0.88622692545275801364908374167057259515

The target is therefore unambiguous: $\sqrt{\pi}/2$.

Python results.

Library · method	Value	Rel. error
'differint.RL'	0.8862272572257357	3.7×10^{-7}
'differint.RLpoint'	0.8862272572257526	3.7×10^{-7}
'differint.CaputoL1point'	0.8862272572033835	3.7×10^{-7}
'numfracpy.RL_der1'	0.8862272572033888	3.7×10^{-7}
'pyfod.riemannliouville'	0.8862329165941121	6.8×10^{-6}
'pyfod.caputo'	0.863550786989199	$** 2.6 \times 10^{-2} **$

Discussion. The four differint/numfracpy Riemann–Liouville and Caputo paths agree to $\sim 3.7 \times 10^{-7}$, the expected accuracy of their second-order trapezoidal / *L1* discretizations. pyfod's Riemann–Liouville route is less accurate (6.8×10^{-6}), consistent with its Gauss–Legendre quadrature approach.

The clear outlier is **pyfod.caputo** at 2.6×10^{-2} . Reading the source, pyfod's Caputo scheme approximates the integrand by a *forward finite difference* of f ,

$$f'(t) \approx \frac{f(t) - f(t - dt)}{dt} \quad (\text{fod.py:228–229}),$$

and then integrates. For $f = \sqrt{x}$ the true derivative $f'(t) = \frac{1}{2\sqrt{t}}$ is singular at the lower limit $t = 0$, so this finite-difference approximation is poor near the endpoint and biases the whole integral. The defect is specific to singular f' , not a general pyfod failure: on smooth test functions (where f' is bounded at 0) the same routine is accurate —

f	f' at $t = 0$	'pyfod.caputo' rel. error
t	1 (constant)	9.7×10^{-7}

t^2	0 (bounded)	2.7×10^{-4}
$\sqrt{t}$	∞ (singular)	2.6×10^{-2}

So pyfod's Caputo operator should be used with care on functions with endpoint-singular derivatives.

4. Test 2 — order three-halves, $\alpha = \frac{3}{2}$

Derivation. With $\beta = \frac{1}{2}$ and $\alpha = \frac{3}{2}$ ($n = 2$), the Riemann-Liouville derivative is the second ordinary derivative of a half-order integral:

$$I^{1/2}\sqrt{x} = \frac{1}{\Gamma(1/2)} \int_0^x (x-t)^{-1/2} t^{1/2} dt.$$

The integral is a beta integral,

$$\int_0^x (x-t)^{-1/2} t^{1/2} dt = x^{1/2+1/2} B\left(\frac{1}{2}, \frac{3}{2}\right) = x \frac{\Gamma(\frac{1}{2})\Gamma(\frac{3}{2})}{\Gamma(2)} = x \cdot \frac{\sqrt{\pi}(\sqrt{\pi}/2)}{1} = \frac{\pi}{2} x,$$

so

$$I^{1/2}\sqrt{x} = \frac{1}{\sqrt{\pi}} \cdot \frac{\pi}{2} x = \frac{\sqrt{\pi}}{2} x,$$

which is **linear** in x . Hence

$$D_{RL}^{3/2}\sqrt{x} = \frac{d^2}{dx^2} \left(\frac{\sqrt{\pi}}{2} x \right) = 0.$$

Equivalently, the power rule gives $\Gamma(\beta+1)/\Gamma(\beta+1-\alpha) = \Gamma(\frac{3}{2})/\Gamma(0) = 0$ (the Gamma pole). The target is therefore exactly **zero**, and the metric is absolute error.

A related point: the **Caputo** derivative of order $\frac{3}{2}$ is *undefined* for $\sqrt{x}$, because it requires $f'' \in L^1$, while $f''(x) = -\frac{1}{4x^{3/2}}$ is not integrable at the origin. Any finite or infinite output from a "Caputo 3/2" routine on this function is, by construction, not a meaningful value.

Wolfram reference.

Reference route	Value
Closed form (Gamma pole)	0
<i>SymbolicR-L</i> ($I(x) = \frac{\pi x}{2} \text{linear} \Rightarrow 2\text{ndderiv} = 0$)	0
Numeric R-L (2nd central difference of I)	-1.4×10^{-21}

Direct Grünwald-Letnikov sum ($M = 5000$)	-4.9782×10^{-7}
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The three analytic/numeric R-L routes confirm the zero target to machine precision; the direct Grünwald-Letnikov sum lands at -4.98×10^{-7} , the expected finite-memory truncation of the semi-infinite sum (see below).

Python results.

Correct — agree with the Wolfram Grünwald-Letnikov value to $\sim 5 \times 10^{-7}$:

Method	Value	Note
'pyfod.grunwaldletnikov'	-4.9786×10^{-7}	slow (75 s; $O(M^2)$ symbolic loop)
'differint.GLpoint'	-6.960×10^{-7}	fast, pointwise
direct Grünwald-Letnikov sum (this work)	-4.9765×10^{-7}	independent cross-check

Slow convergence (valid L2 scheme, low order for singular f):

Method	Value @ $dt = 10^{-4}$	Convergence in dt
'numfracpy.RL_der2'	3.99×10^{-3}	$3.85 \times 10^{-2} \rightarrow 1.26 \times 10^{-2} \rightarrow 3.99 \times 10^{-3} \rightarrow 1.30 \times 10^{-3} \text{ for } dt = 10^{-2} \rightarrow 10^{-3}, \text{ then rounds off below } \sim \text{eps}10^{-6}$

Wrong or unusable at $\alpha = \frac{3}{2}$:

Method	Behavior
'differint.GL' (FFT)	0.465 at $x = 0.5$ — circular-convolution artifact; wrong even at $\alpha = \frac{1}{2}$ (gives 0.679 vs 0.886)
'differint.RL', 'differint.RLpoint'	crash — 'ZeroDivisionError: zero to a negative power' ('RLcoeffs' at 'index_j=0', $\alpha > 1$)

'pyfod.riemannliouville'	'NaN' (documented for $\alpha < 1$; divide-by-zero in quadrature)
'numfracpy.RL_der1'	rejects $\alpha \geq 1$ (returns a string)
'differint.CaputoL1point'	<i>rejects</i> $\alpha \notin (0, 1)$
'pyfod.caputo'	$-\infty$ (finite-diffs the singular f')
'numfracpy.Caputo_der2'	-79.78 (Caputo $\frac{3}{2}$ is mathematically undefined here)

On the $\sim 5 \times 10^{-7}$ Grünwald-Letnikov residual. This is *not* a method error. The Grünwald-Letnikov sum is semi-infinite; any implementation truncates it at $m = \lfloor x/h \rfloor$, and the discarded tail contributes a small, systematic offset. Crucially, the offset is consistent across three independent implementations (pyfod, differint, and the direct sum written for this work) and across the Wolfram reference, which is the signature of a shared truncation effect rather than a bug. The differint.GLpoint value converges toward it as the grid is refined:

'differint.GLpoint'	Value
N	
1001	-5.58×10^{-6}
2001	-1.97×10^{-6}
4001	-6.96×10^{-7}
8001	-2.46×10^{-7}

5. Comparison of the three libraries

A support-and-accuracy matrix across the two tests:

Library	Method	α range	T1 ($\alpha = \frac{1}{2}$)	T2 ($\alpha = \frac{3}{2}$)
'differint'	'RL' (matrix)	(0, 1)	$\checkmark 3.7 \times 10^{-7}$	$\square$ crash
'differint'	'RLpoint'	(0, 1)	$\checkmark 3.7 \times 10^{-7}$	$\square$ crash
'differint'	'CaputoL1point'	(0, 1)	$\checkmark 3.7 \times 10^{-7}$	$\square$ rejects

'differint'	'GLpoint'	$any\alpha$	—	$\checkmark 7.0 \times 10^{-7}$
'differint'	'GL' (FFT)	$any\alpha$	$\square$ wrong (0.679)	$\square$ wrong (0.465)
'pyfod'	'riemannliouville'	$[0, 1)$	$\checkmark 6.8 \times 10^{-6}$	$\square$ 'NaN'
'pyfod'	'caputo'	$[0, 1)$	$2.6 \times 10^{-2}(\text{singular } f')$	$-\infty$
'pyfod'	'grunwaldletnikov'	$\alpha \geq 1$	—	$\checkmark 5.0 \times 10^{-7}(75s)$
'numfracpy'	'RL_der1'	$(0, 1)$	$\checkmark 3.7 \times 10^{-7}$	$\square$ rejects
'numfracpy'	'RL_der2'	$(1, 2)$	n/a	$slow(4.0 \times 10^{-3})$
'numfracpy'	'Caputo_der2'	$(1, 2)$	n/a	$\square$ undefined

Findings.

1. **Only the Grünwald-Letnikov paths handle $\alpha \in (1, 2)$ for this function**, and they agree with the Wolfram reference to $\sim 5 \times 10^{-7}$. `differint.GLpoint` is the recommended choice in this regime: fast, pointwise, and correct.
2. **`differint.GL`, the FFT-based full-array routine, is defective.** It performs a *circular* convolution rather than the causal one the Grünwald-Letnikov definition requires, and it is wrong at every order tested — 0.679 against 0.886 at $\alpha = \frac{1}{2}$, and 0.465 (with a spurious 8.05 near $x = 0.1$) at $\alpha = \frac{3}{2}$. This is a genuine bug worth reporting upstream; the pointwise `GLpoint` routine in the same package is correct.
3. **The Riemann-Liouville-specific routines all break at $\alpha > 1$:** `differint.RL/RLpoint` crash on a zero-to-negative-power in the coefficient kernel, `pyfod.riemannliouville` returns NaN, and `numfracpy.RL_der1` refuses the input. `numfracpy.RL_der2` covers $(1, 2)$ but converges slowly (roughly $O(dt^{0.6})$ here) and hits roundoff by $dt = 10^{-6}$.
4. **`pyfod.caputo` degrades on endpoint-singular f'** , because it finite-differences the integrand. It is accurate on smooth functions and should be avoided for $\sqrt{x}$ -type singularities.
5. **The small $\sim 5 \times 10^{-7}$ Grünwald-Letnikov residual in T2 is finite-memory truncation**, not error: it is consistent across three independent implementations and the Wolfram reference, and it shrinks as the grid is refined.

6. Conclusion

The experiment used two analytically tractable test problems — a nonzero target ($\sqrt{\pi}/2$) and an exact null-space target (0) — to audit three Python fractional-calculus libraries against a four-way Wolfram reference. The reference proved decisive: it fixed the targets to 20 digits and exposed both the successes and the failures of the libraries.

`differint`, `pyfod`, and `numfracpy` are each **correct and accurate within their documented order ranges** for the half-order derivative, agreeing with the reference to 10^{-7} - 10^{-6} . Beyond that range the picture is more uneven. For $\alpha \in (1, 2)$ only the Grünwald-Letnikov routines are usable, `differint.GLpoint` being the standout; the FFT-based `differint.GL` is incorrect at every order; and the Riemann-Liouville and Caputo routines either crash, return NaN, or — in the Caputo $\frac{3}{2}$ case — evaluate an operator that is mathematically undefined for $\sqrt{x}$. The single most important practical takeaway: **match the routine's order range to your α , prefer the Grünwald-Letnikov paths for $\alpha \geq 1$, and treat `differint.GL` and `pyfod.caputo` on singular- f' with suspicion.**

Artifacts (in C:\Users\MATTIA\source\hermes-dir\): frac_verify.py / frac_verify.wsc / frac_results.json (T1) and frac_verify_15.py / frac_verify_15.wsc / frac_results_15.json (T2). Environment: Python 3.14.3 (venv), WolframScript 1.13.0 / Mathematica kernel 14.2.1.